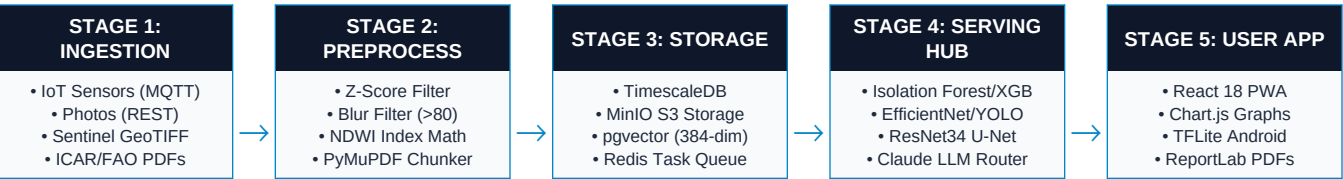


# Aquaculture Intelligence System (AIS)

## End-to-End System Architecture & Technical Workflow Flowcharts

|                          |                                                                    |
|--------------------------|--------------------------------------------------------------------|
| System Design Pattern:   | Layered Microservices & Event-Driven Ingestion Gateway             |
| Primary Storage Engine:  | PostgreSQL 16 + TimescaleDB Hypertables + pgvector HNSW + MinIO S3 |
| Model Serving Framework: | FastAPI + ONNX Runtime (Cloud) & TFLite FP16 (Android Offline)     |
| Gen AI Orchestrator:     | Claude API Sonnet 4-6 with Parallel Tool Calling & RAG Retrieval   |

### 1. High-Level System Architecture Flowchart



### 2. Module-Specific Execution Flowcharts

#### Module 1: Water Quality Monitoring & 72-Hour Forecasting Workflow

**WORKFLOW:** IoT Sensors (DO, pH, Temp, NH3) ■■■> MQTT Topic 'pond/{id}/sensors' ■■■> Pydantic v2 Range Validation ■■■> Z-Score Spike Filter ( $|Z| > 3.5$ ) ■■■> TimescaleDB Hypertable 'sensor\_readings' (7-day chunks) ■■■> Isolation Forest Anomaly Scoring & XGBoost Regressors (24h/48h/72h forecasts) ■■■> WebSocket Push & Stress Risk Badge (Low/Med/High/Critical)

#### Module 2: Fish Disease & Stress Detector Workflow

**WORKFLOW:** Farmer Smartphone Photo ■■■> REST Multipart Upload ■■■> MinIO S3 Object Store ('images/raw/{uuid}.jpg') ■■■> OpenCV Laplacian Blur Filter (blur\_score >= 80 check) ■■■> YOLOv8 Net-Pull Fish Batch Crop ■■■> PyTorch EfficientNet-B2 Classifier (15 classes) via ONNX Runtime (Cloud) / TFLite FP16 (Offline Android) ■■■> ICAR-CIFA Knowledge Base Lookup (Treatment, Withdrawal Period) ■■■> Disease Diagnosis UI Card

#### Module 3: Fish Growth & Yield Forecaster Workflow

**WORKFLOW:** Farmer Stocking Record Input (Species, Stocking Weight, Count) ■■■> ICAR-CIFA SGR Temperature Lookup Matrix ■■■> Bioenergetic Biomass Engine [ $W = W_0 * e^{(SGR * days)}$ ] ■■■> Environmental Temp/DO Fetch ■■■> XGBoost Growth Forecaster (t+7, t+14, t+30 days) ■■■> Binary Search Target Harvest Date Predictor ( $\pm 5$  days) ■■■> Logistic Regression Mortality Risk Score ■■■> Interactive Growth Curve Chart

#### Module 4: Feed Optimization Engine Workflow

**WORKFLOW:** Current Biomass Estimate (Growth Module) ■■■> Environmental Water Temp & DO Read (TimescaleDB) ■■■> Thermal & Oxygen Feeding Correction (-15-20% per 3°C below optimal; -30% if DO < 4 mg/L) ■■■> XGBoost Daily Feed Regressor (kg/day) ■■■> Dawn/Dusk Schedule Optimizer (60% Morning / 40% Evening) ■■■> FCR Drift Tracker (Alert if FCR > 1.15x target for 7d) ■■■> Feed Quantity Recommendation Card

#### Module 5: Generative AI Advisory & Report Engine Workflow

**WORKFLOW:** Farmer Voice / Text Query ■■■> Whisper ASR Voice Transcription ■■■> sentence-transformers Embedding (384-dim) ■■■> pgvector HNSW Cosine Similarity Search (Top-5 chunks from 2,000-page ICAR/FAO corpus) ■■■> Claude API Orchestrator (Sonnet 4-6) + Function Calling (get\_water\_quality, predict\_disease) ■■■> SSE Streaming Text Output + Speech Synthesis (Google TTS/Bhashini) + ReportLab PDF Generator

Module 6: Drone/Satellite Pond Monitor Workflow

**WORKFLOW:** Celery Beat Scheduled Task  Sentinel Hub API Query (<20% cloud filter)  Fetch 6 Sentinel-2 L2A Bands (B02, B03, B04, B08, B8A, B11)  Compute NDWI Mask = (B03-B08)/(B03+B08) & Chlorophyll-a Proxy = B03/B04  ResNet34 U-Net Land Cover Segmentation  Tile Caching in MinIO S3  React Leaflet.js Interactive Satellite Map with Layer Controls

3. End-to-End Data Protocol & Latency Budget Table

| Workflow Step             | Data Input Format              | Protocol / Channel    | Target Latency SLA   | Storage Target         |
|---------------------------|--------------------------------|-----------------------|----------------------|------------------------|
| IoT Telemetry Ingestion   | JSON (do, ph, temp, nh3)       | MQTT v5 (Port 1883)   | < 100 ms             | TimescaleDB Hypertable |
| Water Quality Forecast    | 15m/1h/6h Feature Vector       | HTTP REST / WebSocket | < 200 ms             | prediction_logs Table  |
| Disease Image Inference   | RGB Photo (224x224x3)          | HTTP REST / ONNX      | < 120 ms (Cloud CPU) | MinIO S3 Blob Storage  |
| Offline Disease Inference | RGB Photo (224x224x3)          | TFLite FP16 (Android) | < 200 ms (Mobile)    | Local IndexedDB Cache  |
| Growth & Harvest Date     | Biomass & Environmental Vector | HTTP REST             | < 150 ms             | pond_cycles Table      |
| Daily Feed Optimization   | Biomass & Temp/DO Vector       | HTTP REST             | < 100 ms             | feed_logs Table        |
| Gen AI Advisory Chat      | Text / Audio Stream            | HTTP SSE / Voice API  | < 3.0 sec (P95)      | rag_chunks (pgvector)  |
| Satellite Scene Render    | 6-Band GeoTIFF (10m)           | Sentinel Hub REST API | < 20 min (Batch)     | Leaflet.js GeoJSON Map |